Smart Traffic Light System

Architecture Overview

1. Input Layer: Real-Time Camera Feed

1. Multiple CCTV cameras installed at intersections provide live video streams.
2. Video streams are fed into the processing system.

2. Data Ingestion and Preprocessing

1. Video frames are extracted at a fixed rate (e.g., 1 frame per second).
2. Preprocessing includes resizing, normalization, and possibly ROI (Region of Interest) cropping to focus on the road/lane area.

3. Traffic Density Estimation Module (AI Model)

1. Use computer vision models to detect and count vehicles.
2. Possible models:
3. Object detection (YOLOv5/YOLOv8, Faster R-CNN) to detect cars, bikes, trucks.
4. Alternatively, segmentation models to estimate density.
5. Output: Number of vehicles or traffic density metric per lane.

4. Traffic Analysis and Decision Engine

1. Aggregates counts from multiple lanes.
2. Applies a traffic control algorithm (rule-based or reinforcement learning) to decide optimal light timing.
3. Algorithm considers:
4. Current vehicle density.
5. Historical traffic patterns.
6. Priority (emergency vehicles, pedestrians).

5. Traffic Signal Controller

1. Sends commands to traffic lights to adjust timings dynamically.
2. Can interface with existing traffic light control hardware via APIs or controllers.

6. Monitoring & Dashboard

1. Real-time dashboard to visualize traffic density, camera feeds, and signal status.
2. Alerts for anomalies or system errors.